Design and Calibration of a Digital Range Finder using ESP32

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***Abstract*—A non-contact digital range finder was developed using an ESP32 microcontroller and an ultrasonic transducer. The instrument generates short bursts of ultrasound, measures the echo time of flight (ToF) and converts this to distance. A real-time distance (10–200 cm) is displayed on an I²C liquid crystal display (LCD). Because the speed of sound depends on temperature and, to a lesser extent, humidity, this dependence is discussed in the measurement model. Signal processing techniques such as moving-average filtering and Kalman filtering were explored to reduce measurement noise and improve accuracy. Calibration was carried out by comparing the sensor's output with a standard measuring tape at various distances, and performance metrics including linearity and error percentage were computed.**

I. Introduction

Accurate distance measurement is central to many electrical and electronic engineering applications, ranging from robotics and industrial automation to fluid-level control and non-contact gauging. Ultrasonic rangefinders provide a low-cost, robust method to make such measurements because they are insensitive to ambient light and electromagnetic interference [1]. These sensors operate by emitting a short burst of high-frequency sound and measuring the time elapsed until an echo returns from a target. Non-contact distance measurement avoids mechanical wear and is safer for applications involving moving parts or hazardous environments.

The ESP32 microcontroller family integrates a dual-core 32-bit processor, Wi-Fi, Bluetooth and a rich set of peripherals including UART, SPI, I²C and ADC interfaces. These modules operate from 3.0–3.6 V, have up to 26 GPIO pins and can clock at up to 240 MHz. Their high computing power and low power consumption make them suitable as the control unit for a digital range finder. An I²C LCD provides a simple user interface to display the measured distance [2].

This project aims to design and build a digital range finder that uses ultrasonic time-of-flight measurement with the ESP32 and to calibrate the device over a 10–100 cm range. The report describes the physics of ultrasonic ranging, factors affecting measurement accuracy and advanced signal-processing techniques, then outlines an action plan for implementing and calibrating the instrument.

II. Measurement Principle and Mode

Ultrasonic ranging can be implemented using several methods, including pulse-echo time-of-flight (ToF), phase-shift-based measurement, and frequency-modulated/chirp-based ranging approaches [1]. Each method has different trade-offs in hardware complexity, processing requirements, and noise robustness.

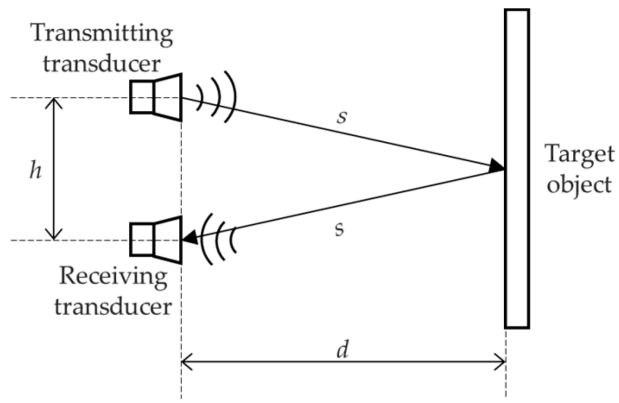

Fig. 1. Pitch-catch ultrasonic ranging geometry (adapted from [1]).

In this project, we employed the ToF method because of its simple implementation with the ESP32 platform. The system was configured in a pitch-catch mode, where separate transmitting and receiving transducers are used.

A. Time-of-Flight Principle

For a generic pulse-echo ToF configuration, distance is computed from travel time as

$$d = \frac{c\,t}{2} \tag{1}$$

where c is the speed of sound in air and t is the measured round-trip time.

For the pitch-catch geometry used in this work, Fig. 1 shows two separated transducers with spacing h. If the total ToF is t, then each acoustic path has length

$$s = \frac{c\,t}{2} \tag{2}$$

Using the right-triangle relation in the figure, the forward standoff distance to the target is

$$d = \sqrt{s^2 - \left(\frac{h}{2}\right)^2} = \sqrt{\left(\frac{c\,t}{2}\right)^2 - \left(\frac{h}{2}\right)^2} \tag{3}$$

When $h \ll 2d$, this reduces approximately to $d \approx \frac{c\,t}{2}$.

B. Speed of Sound in Air and Environmental Compensation

The speed of sound in air depends primarily on the ambient temperature. In dry air at $0°$C, the speed of sound is approximately

$$c \approx 331.45\ \mathrm{m\,s^{-1}} \tag{4}$$

A commonly used linear approximation relating sound speed to air temperature is [1]

$$c \approx 331.45 + 0.607\,T_C \tag{5}$$

where T_C is the air temperature in degrees Celsius. This relation indicates that the speed of sound increases by about $0.6\,\mathrm{m\,s^{-1}}$ for every $1°$C increase in temperature.

Environmental factors such as humidity and atmospheric pressure also affect the propagation speed of sound, though their influence is comparatively smaller. Across typical ranges of 0–100% relative humidity and 50–110 kPa pressure, the variation in sound speed is generally less than 0.7% [1].

For improved measurement accuracy, the ESP32 can update distance calculations using a temperature-dependent value of c. If ambient temperature is known, systematic errors caused by environmental variations can be reduced.

III. Methodology and Design

A. Hardware Implementation

The prototype consists of the following main components:

- ESP32 development board
- HC-SR04 ultrasonic sensor
- Low-power laser module mounted at the centerline of the ultrasonic transducers
- I^2C 16×2 LCD module with backpack
- Breadboard, jumper wires, and regulated power supply

1) Circuit Connections: The ESP32 development board is powered using a regulated 3.3 V supply. The ultrasonic sensor module is powered from the ESP32 supply rail, and its ground is connected to the ESP32 ground to establish a common reference.

The Trigger pin of the ultrasonic sensor is connected to a digital GPIO pin configured as an output. The Echo pin is connected to a GPIO pin configured as an input.

The I^2C LCD module is connected to the ESP32 I^2C interface. The SDA and SCL lines are connected to GPIO 21 and GPIO 22 respectively.

The laser module is mechanically fixed so that its beam emerges from the middle of the two ultrasonic transducers. This laser is used only as a visual pointer to align the measurement axis with the target point.

2) Prototype Assembly: All components are assembled on a breadboard during the prototyping stage. The ultrasonic transducer is positioned so that it has a clear line of sight to the target surface. The laser pointer is adjusted during assembly so its spot coincides with the center of the ultrasonic sensing direction. Care is taken to minimize nearby reflective objects that could introduce unwanted echoes. If necessary, simple shielding or damping materials can be placed around the sensor to reduce acoustic reflections from surrounding surfaces.

B. Software Flow

The ESP32 generates a trigger pulse sequence for each measurement cycle. First, the trigger pin is set LOW for $2\,\mu$s, then set HIGH for $10\,\mu$s, and then returned LOW. The echo pulse width is measured using the Arduino `pulseIn()` function on the echo pin with a timeout of $30{,}000\,\mu$s. The detailed software and firmware implementation used in this project is available in the project repository [3].

If no valid echo pulse is received within the timeout, that cycle is ignored. If a valid pulse is received, the time value is stored and the distance is computed using Eq. (3). The resulting distance is then sent to the active user interface mode.

Two software operating modes are implemented:

- **LCD mode (local mode):** after powering the ESP32, pressing the BOOT button once starts local display mode, where distance is shown on the LCD.
- **Website mode (Wi-Fi mode):** after reset, pressing the BOOT button twice starts web mode. The ESP32 starts the web interface and displays its IP address. A user connects over Wi-Fi, opens the IP address in a browser, and can view live readings, record readings, and perform calibration tasks [3].

IV. Calibration and Results

A. Calibration Procedure

The developed firmware includes a web-based calibration interface for updating the ranging parameters of the ESP32-based measurement system [3]. After powering the device, the ESP32 can be placed into Wi-Fi mode by pressing the boot button twice. In this mode, the user connects to the firmware-hosted local network interface and accesses the calibration page through a web browser.

For calibration, the sensor is positioned at a known distance from a target object, measured beforehand using a reference scale. The same reference distance is then entered into the firmware interface, and the calibration command is executed. During this process, the transmitter emits an ultrasonic pulse and the system measures the corresponding time-of-flight from the received signal. Using the known reference distance and the measured travel time, the firmware estimates an effective sound velocity for the current operating condition.

This procedure is important because ultrasonic propagation in air is affected by environmental conditions, particularly temperature and humidity. In addition, practical measurement error may arise from sensor characteristics, alignment imperfections, and hardware-related timing offsets. By calibrating the device at a known distance, these systematic effects can be partially compensated, allowing the firmware to adjust its distance calculation for improved accuracy under the prevailing conditions.

TABLE I
READINGS & CORRESPONDING ERRORS WHEN DATA IS PROCESSED IN PULSE-ECHO MODE

Actual (cm)	Raw reading (cm)	Raw reading with moving avg (cm)	Raw error (%)	Moving avg error (%)
1.5	1.71	1.71	14.00	14.00
2	1.71	1.71	14.50	14.50
3	1.71	2.76	43.00	8.00
4	3.99	3.99	0.25	0.25
5	4.96	5.56	0.80	11.20
6	4.65	4.65	22.50	22.50
7	7.92	7.72	13.14	10.29
8	7.92	7.91	1.00	1.12
9	8.92	8.92	0.89	0.89
10	9.96	9.95	0.40	0.50
15	15.44	15.46	2.93	3.07
20	19.47	19.47	2.65	2.65
25	24.59	24.6	1.64	1.60
30	29.68	29.69	1.07	1.03
35	34.29	34.61	2.03	1.11
40	40.07	40.06	0.18	0.15
45	44.52	44.53	1.07	1.04
50	50.53	50.2	1.06	0.40
55	54.86	54.54	0.25	0.84
60	59.77	59.68	0.38	0.53
70	69.51	53.31	0.70	23.84
80	79.13	79.12	1.09	1.10
90	89.38	87.18	0.69	3.13
100	99.45	99.47	0.55	0.53
120	120.48	120.47	0.40	0.39
140	140.21	140.24	0.15	0.17
148	148.99	149.27	0.67	0.86
165	166.01	166.5	0.61	0.91

TABLE II
READINGS & CORRESPONDING ERRORS WHEN DATA IS PROCESSED IN PITCH-CATCH MODE

Actual (cm)	Pitch-catch (cm)	Pitch-catch with moving avg (cm)	Pitch-catch error (%)	Pitch-catch moving avg error (%)
1.5	1.39	1.39	7.33	7.33
2	1.39	1.39	30.50	30.50
3	1.39	2.56	53.67	14.67
4	3.87	3.87	3.25	3.25
5	4.86	5.47	2.80	9.40
6	4.54	4.54	24.33	24.33
7	7.85	7.65	12.14	9.29
8	7.85	7.85	1.88	1.88
9	8.87	8.87	1.44	1.44
10	9.91	9.9	0.90	1.00
15	15.41	15.42	2.73	2.80
20	19.44	19.44	2.80	2.80
25	24.57	24.58	1.72	1.68
30	29.67	29.67	1.10	1.10
35	34.28	34.6	2.06	1.14
40	40.06	40.05	0.15	0.12
45	44.51	44.52	1.09	1.07
50	50.52	50.19	1.04	0.38
55	54.85	54.53	0.27	0.85
60	59.76	59.67	0.40	0.55
70	69.5	53.3	0.71	23.86
80	79.13	79.11	1.09	1.11
90	89.38	87.18	0.69	3.13
100	99.45	99.46	0.55	0.54
120	120.47	120.47	0.39	0.39
140	140.21	140.24	0.15	0.17
148	148.99	149.27	0.67	0.86
165	166	166.5	0.61	0.91

B. Data and Analysis

The sensor readings were processed in both pulse-echo mode and pitch-catch mode in order to evaluate and compare their performance.

In addition to the distance calibration results, the required clearance along the acoustic path was also investigated. This was done to ensure that the sensor measured the intended target distance rather than detecting unwanted objects located near the propagation path or within the peripheral region outside the direct line of measurement.

TABLE III
REQUIRED CLEARANCE AT DIFFERENT MEASUREMENT INTERVALS.

Distance range (cm)	Required clearance (cm)
0–30	7
30–60	26
60–90	36
90–120	45
120–150	65

C. Performance Graph

The following plots summarize the measured sensor response, the error trend across distance, the average error in different operating ranges, and the clearance requirement.

Figure 2 shows that, over most of the measured range, the processed sensor outputs follow the ideal distance-response line closely. The agreement is strongest in the medium and far ranges, where the plotted points remain near the reference line for all four processing cases. Larger deviations are mainly observed in the close range, especially below approximately 10 cm, where the measured points are more scattered and the relative error becomes significantly higher. This behavior is also consistent with the tabulated results, where the shortest-distance measurements produce the largest percentage errors.

TABLE IV
MEAN ABSOLUTE ERROR DIFFERENT DISTANCE RANGES AND PROCESSING METHODS.

Range (cm)	Method	MAE before	MAE after outlier removal
All (0–200)	Raw reading (pulse echo) with moving avg	1.128571	0.465000
	Raw reading (pulse echo)	0.478214	0.478214
	Pitch-catch with moving avg	1.154643	0.492692
	Pitch-catch	0.514286	0.514286
Close (0–10)	Raw reading (pulse echo) with moving avg	0.360000	0.360000
	Pitch-catch with moving avg	0.425000	0.425000
	Raw reading (pulse echo)	0.431000	0.431000
	Pitch-catch	0.528000	0.528000
Far (60–200)	Raw reading (pulse echo)	0.605556	0.605556
	Pitch-catch	0.605556	0.605556
	Raw reading (pulse echo) with moving avg	2.746667	1.003750
	Pitch-catch with moving avg	2.751111	1.007500
Medium (10–60)	Raw reading (pulse echo) with moving avg	0.331818	0.331818
	Pitch-catch with moving avg	0.340909	0.340909
	Raw reading (pulse echo)	0.354545	0.354545
	Pitch-catch	0.363636	0.363636

The error curves in Figure 3 indicate that the main performance limitation is concentrated at short distances and at a few isolated outlier points. In the full-range plot, most data beyond the close region remain within a low error band, generally around 1–3%, whereas very large peaks appear at the smallest measured distances and at one anomalous point in the far range when moving average processing is applied. The medium-range plot shows the most stable overall response,

with comparatively smooth behavior and small deviations for all methods. In contrast, the far-range plot suggests that the raw pulse-echo and raw pitch-catch results remain relatively stable, while the moving-average cases are more sensitive to occasional abnormal measurements.

The mean absolute error comparison in Figure 4 and Table IV further clarifies this trend. After outlier removal, the best overall performance across the full range is obtained using raw pulse-echo readings with moving average, which gives an MAE of 0.465 cm. In the close range, the same method also produces the smallest MAE of 0.360 cm, while in the medium range it again provides the lowest value of 0.3318 cm. However, in the far range, the unfiltered raw pulse-echo and raw pitch-catch methods both perform better, each yielding an MAE of 0.6056 cm, whereas the moving-average methods rise to approximately 1.0 cm after outlier removal. This indicates that averaging is beneficial in short and medium distances, but it may amplify the effect of occasional abnormal samples in long-range measurements. From a practical perspective, the full-range average error of approximately 0.45–0.50 cm suggests that the system is more suitable for applications that can tolerate about ± 0.45 cm uncertainty than for precision-critical measurement tasks.

Figure 5 complements the distance results by illustrating the required clearance around the measurement axis. The clearance increases with distance, from 7 cm in the 0–30 cm interval to 65 cm in the 120–150 cm interval. This trend is expected because the ultrasonic beam spreads as distance increases, enlarging the region from which reflections may be received. Consequently, accurate long-range measurement requires a wider unobstructed path to avoid detecting nearby peripheral objects instead of the intended target. The clearance plot also indicates that the theoretical measurement angle of approximately 15° reported in the HC-SR04 datasheet [4] agrees reasonably well only up to about 30 cm. Beyond this region, the experimentally observed boundary rises to about 22°, indicating a wider effective beam in practice than in the simplified theoretical model.

V. Error Analysis and Discussion

The main sources of error in this work are likely related to measurement setup, environmental variation, and sensor behavior. Larger deviations appear mainly at short distances and at a few isolated outlier points, suggesting that the error is not caused by one factor alone.

One possible source is human error during measurement and calibration. Small inaccuracies in setting the reference distance, reading the scale, or entering the calibration value into the firmware may affect the calculated correction factor. Parallel alignment error between the sensor and the target surface may also influence the time-of-flight by changing the effective reflection path.

Environmental conditions may also contribute to the observed differences. Since the speed of sound depends on temperature and humidity, any variation in the surrounding air can change the measured distance. In addition, unwanted reflections, acoustic disturbance, and nearby obstacles may introduce noise into the received signal.

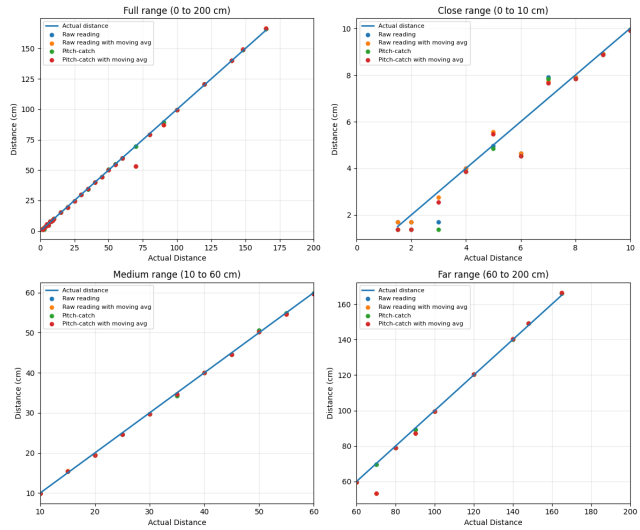

Fig. 2. Sensor reading plots across different distances.

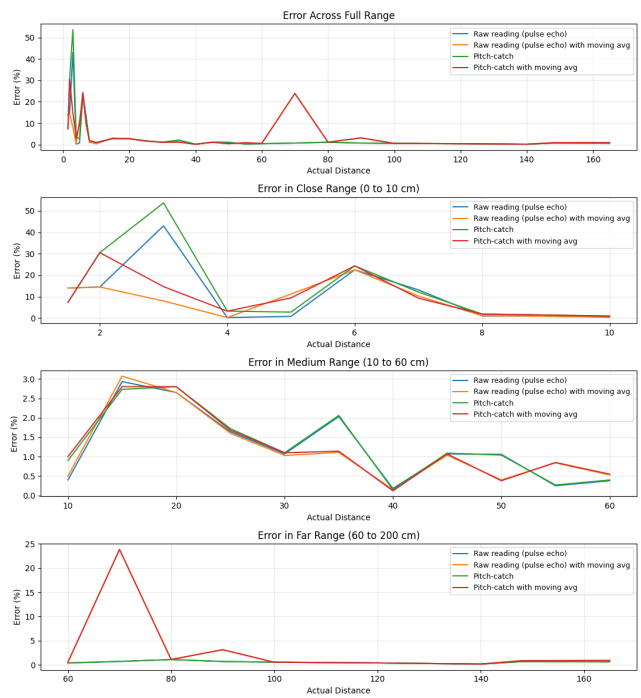

Fig. 3. Error plots across different distances for the measured data.

Sensor and circuit limitations may be another reason for the remaining error. Timing jitter, inconsistent echo detection, or reduced response stability at some distances may produce abnormal samples. This may explain why some moving-average results become worse when outliers are present.

The clearance results also suggest a practical geometric limitation. Although the system was able to detect distance up to 165 cm, the required clearance increases with distance because of beam spreading. A theoretical measurement angle

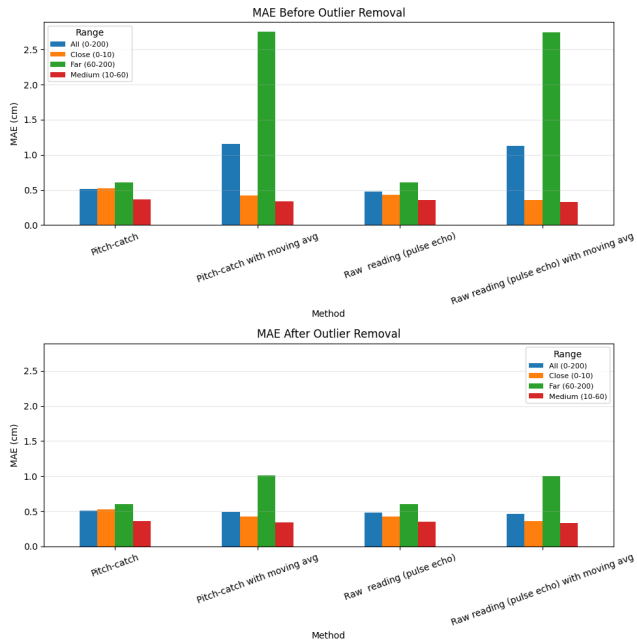

Fig. 4. Mean absolute error at different distance ranges for selecting the best measurement method.

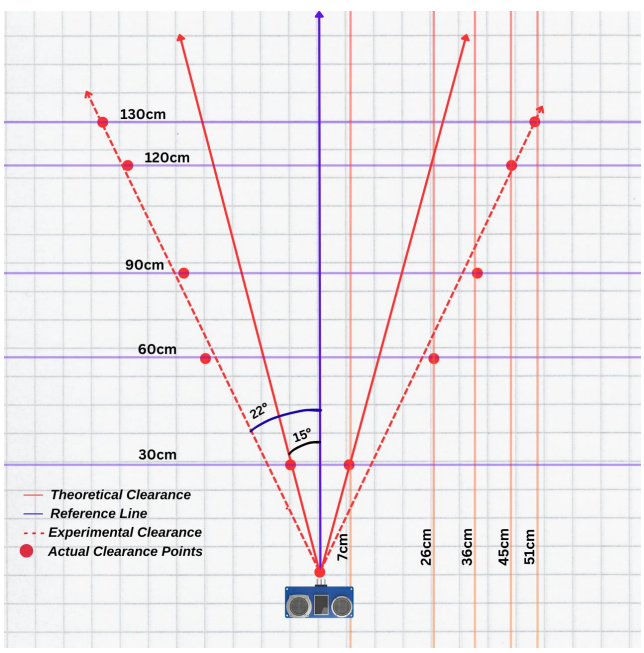

Fig. 5. Measurement angle and clearance plot

of about 15° appears to be reasonable only within the first 30 cm, while the experimental pattern after that region is closer to about 22°. If a maximum allowable clearance of 30 cm is imposed in the path of flight, the practical usable range would be limited to approximately 90 cm even though the sensor can still detect targets beyond that distance.

Thus, the observed errors may be associated with a combination of human measurement uncertainty, parallel alignment error, environmental effects, and sensor noise. Other factors may also be present in the experimental setup. For this reason, the present system may be less appropriate for precision-critical applications, while remaining practically useful in tasks where an uncertainty of roughly ± 0.45 cm is acceptable.

VI. Conclusion

An ESP32-based ultrasonic digital range finder was designed, implemented, and calibrated successfully. The results show that both pulse-echo and pitch-catch processing can provide useful distance measurements, with better consistency in the medium and far ranges than at very short distances. Although accurate measurements were achieved up to 165 cm, the clearance study suggests that, under a practical maximum clearance constraint of 30 cm, the effective usable range is about 90 cm. The observed average tolerance indicates that the system is more suitable for applications with about ± 0.45 cm allowable error than for precision-critical measurement tasks. Nevertheless, the prototype shows good potential for applications such as object detection and object trajectory estimation. Future improvement may be achieved through better sensor alignment, more stable mounting, and stronger filtering or outlier-handling methods.

References

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